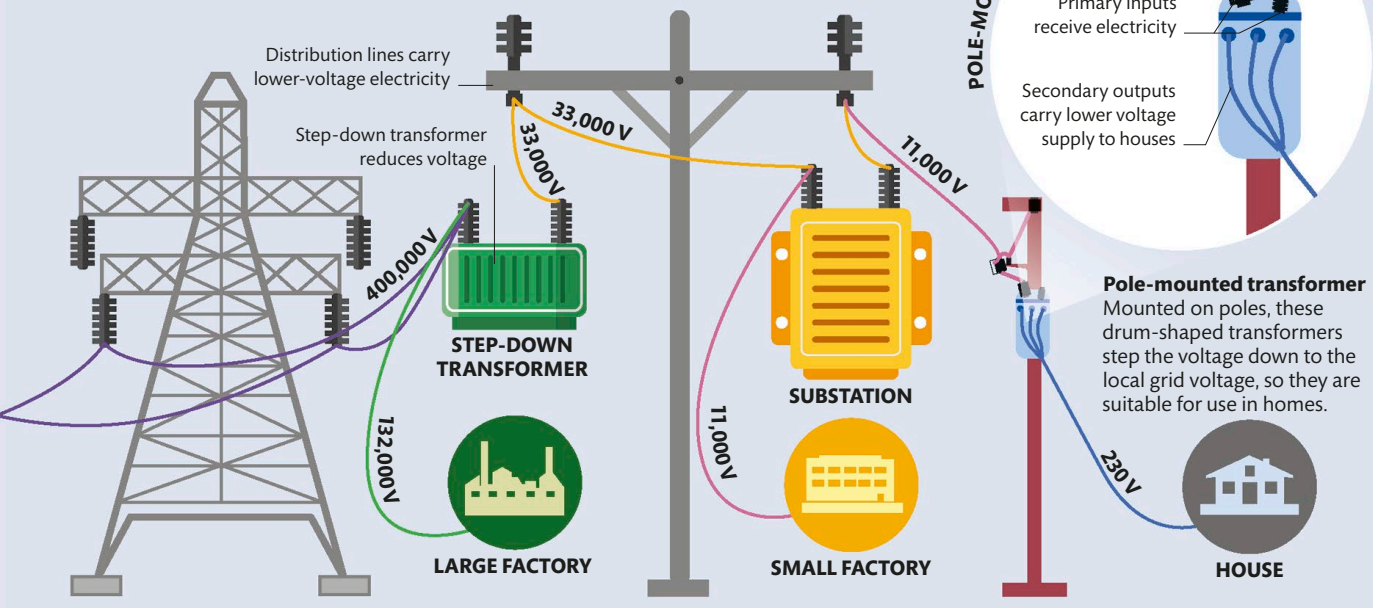
## THE WORLD'S **TALLEST ELECTRICITY PYLON** LINE, IN CHINA, IS **1,200 FT (370 M) HIGH**




**Pole-mounted transformer**
Mounted on poles, these drum-shaped transformers step the voltage down to the local grid voltage, so they are suitable for use in homes.

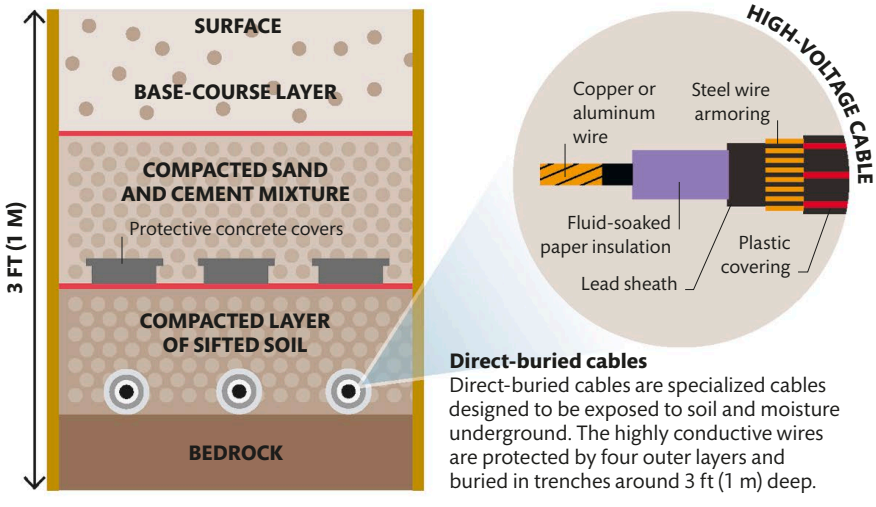
**4 Direct supply to industry**
Some factories with high electricity requirements may take power directly from the high-voltage lines. Other factories require a transformer to step the voltage down to approximately 132,000 volts.

**5 Distribution substation**
The high-voltage electricity is reduced to a much lower voltage at a substation, which usually includes several transformers. From here, it is supplied to smaller industrial and commercial customers.

**6 Domestic supply**
A network of distribution lines supplies electricity to houses. The final voltage reduction takes place at pole-mounted transformers, before the electricity supply passes into a home's fuse box.

## Underground cables

To reduce the visual impact and land use of rows of towers and the power lines they support, many electricity supply cables are buried underground. These cables require multiple layers of protection and insulation. The cables are placed in trenches. Individual cables can be up to 0.6 miles (1 km) long, and extra reinforcement is added to the trenches at the points where the cables are joined. The cables are guarded by concrete covers that prevent accidental cable cutting.




**Direct-buried cables**
Direct-buried cables are specialized cables designed to be exposed to soil and moisture underground. The highly conductive wires are protected by four outer layers and buried in trenches around 3 ft (1 m) deep.